2017-Summer Paper 1 – Basic Sciences in Urology

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Q1. Discuss anatomy of kidney. Describe role of nephron sparing surgery in small renal mass.

Type: LAQ (25 marks)

Surgical anatomy of the kidney

- **Size and position:** about 11 cm, 125–170 g; T12–L3 on psoas, so the upper pole is more medial and posterior. The right kidney is 1–2 cm lower (liver).
- **Relations:**

Aspect	Right	Left
Superior	Liver, adrenal, 12th rib	Spleen, stomach, adrenal, 11th–12th ribs
Medial/anterior	Duodenum (2nd part), hepatic flexure	Pancreatic tail, splenic vessels, splenic flexure, jejunum
Posterior (both)	Diaphragm, psoas, quadratus lumborum, subcostal nerve	Same

- **Coverings (inside out):** true capsule, perirenal fat, **Gerota fascia**, pararenal fat. Gerota fascia is closed above and laterally but open below. So it contains tumour and abscess, but fluid can track into the pelvis.
- **Hilum (anterior to posterior):** renal vein, artery, pelvis, then the posterior segmental artery.
- **Arteries:** from the aorta at L1–L2; the right passes behind the IVC (inferior vena cava). Five **segmental end arteries** (apical, upper, middle, lower from the anterior division; posterior). With no anastomoses, occlusion causes segmental infarction, but each segment can be resected separately. Then interlobar (columns of Bertin), arcuate, interlobular arteries. Accessory arteries in 25–28%.
- **Brödel line:** an avascular plane between anterior and posterior territories, just behind the lateral border. Incision here bleeds least (anatrophic nephrolithotomy).
- **Veins** anastomose freely, so a segmental vein can be ligated safely. The long left renal vein receives adrenal, gonadal and lumbar veins.
- **Parenchyma:** cortex and 8–18 pyramids; 0.4–1.2 million nephrons per kidney.
- **Collecting system:** papilla, minor calyces (7–9), major calyces, pelvis. Puncture through the **calyceal fornix**, not the infundibulum (Sampaio).
- **Also know:** lymph to para-aortic (left) or paracaval/interaortocaval (right) nodes; sympathetic T8–L1.

Diagram to draw: segmental arteries with the Brödel line on a transverse kidney section.

Small renal mass (SRM)

- **Definition:** an enhancing renal mass ≤4 cm (cT1a), usually incidental.
- **Biology:** 20–30% are benign (oncocytoma, fat-poor angiomyolipoma); most cancers are indolent (<2% metastasise).
- **Rationale:** radical nephrectomy (RN) causes CKD (chronic kidney disease), which raises cardiovascular events and death. PN (partial nephrectomy) gives equal cancer control with better function.

Work-up

- Multiphase CT (computed tomography) or MRI (magnetic resonance imaging); chest imaging; eGFR (estimated glomerular filtration rate) and proteinuria. **Renal mass biopsy** if it may change management (non-diagnostic about 14%).
- **Nephrometry scores** predict complexity and complications:

R.E.N.A.L. component	Scored as
Radius	≤4, 4–7, ≥7 cm
Exophytic/endophytic	≥50% exophytic, <50%, entirely endophytic
Nearness to collecting system	≥7, 4–7, ≤4 mm
Anterior/posterior	Descriptor
Location to polar lines	Polar, crossing, central

- Total: low 4–6, moderate 7–9, high 10–12. **PADUA** and C-index are alternatives.

Indications for nephron-sparing surgery

Category	Situations
Imperative	Solitary kidney, bilateral tumours, RN would mean dialysis
Relative	Contralateral kidney threatened (stone, stenosis, diabetes), hereditary RCC (renal cell carcinoma)
Elective	cT1a with a normal contralateral kidney: **standard of care**

Options (AUA (American Urological Association) 2021)

- **Partial nephrectomy: preferred for cT1a.** Open, laparoscopic or robotic (RAPN, robot-assisted PN) with equal oncology. Steps:
  1. Expose kidney; intraoperative ultrasound marks the tumour.
  2. Hilar control; clamp artery (± vein), selective or off-clamp.
  3. Excise with a thin rim (or enucleate); negative margin is the priority.
  4. Close collecting system and vessels; renorrhaphy (sliding clip); unclamp; drain.
- **Ischaemia:** warm ischaemia traditionally <25–30 min; ice slush if longer is expected. Function depends mainly on **parenchyma preserved**; randomised trials show no lasting benefit of off-clamp.
- **Trifecta:** negative margin, no major complication, warm ischaemia ≤25 min.
- **Thermal ablation** (cryoablation, radiofrequency): masses <3 cm, percutaneous, biopsy first; higher local recurrence, salvaged by repeat ablation.
- **Active surveillance:** <2 cm, elderly or comorbid. Intervene if >3 cm or growth >5 mm a year (DISSRM registry: almost no metastasis).
- **RN** only if all three: high complexity, no CKD or proteinuria, and post-RN eGFR likely >45.

Complications of PN

- Urine leak (stent), bleeding and pseudoaneurysm (angioembolisation), arteriovenous fistula, acute kidney injury, positive margin (0–7%; surveillance), hyperfiltration proteinuria.

Evidence

- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011):** RN vs PN for ≤5 cm tumours: PN better function, similar oncology; overall survival slightly better with RN, but the trial is flawed.
- **Also know:** von Hippel–Lindau: treat at 3 cm, enucleation; 3D models.

Key points to remember:

- Five segmental end arteries; Brödel line avascular; puncture at the fornix.
- SRM ≤4 cm: 20–30% benign; PN is standard, ablation <3 cm, surveillance <2 cm.
- R.E.N.A.L. 4–6 / 7–9 / 10–12; trifecta includes warm ischaemia ≤25 min.
- EORTC 30904: PN better function, similar cancer control.

Q2. Describe Pathophysiology of urosepsis and its management.

Type: LAQ (25 marks)

Definitions

- **Urosepsis:** sepsis whose source is the urogenital tract. It is usually Gram-negative and usually follows obstruction or instrumentation.
- **Sepsis-3 (2016):**

Term	Definition
Sepsis	Life-threatening organ dysfunction from a dysregulated host response to infection: **SOFA (Sequential Organ Failure Assessment) rise ≥2**
qSOFA (quick SOFA)	Respiratory rate ≥22, altered mentation, systolic blood pressure ≤100 mmHg; ≥2 = high risk
Septic shock	Vasopressor for MAP (mean arterial pressure) ≥65 mmHg **and** lactate >2 mmol/L despite fluids

- SIRS (systemic inflammatory response syndrome) criteria were dropped as non-specific.

Causes and risk factors

- **Obstruction with infection:** stone, PUJ (pelviureteric junction) obstruction, tumour, pregnancy. Pyonephrosis, emphysematous pyelonephritis, renal, perinephric and prostatic abscess.
- **Instrumentation:** PCNL (percutaneous nephrolithotomy), ureteroscopy with high irrigation pressure, TURP (transurethral resection of prostate), prostate biopsy, catheters.
- **Host:** diabetes, immunosuppression, old age, neurogenic bladder.
- **Organisms:** *E. coli* (commonest), *Proteus*, *Klebsiella*, *Pseudomonas*, *Enterococcus*; ESBL (extended-spectrum β-lactamase) producers rising; *Candida* in catheterised diabetics.

Pathophysiology

1. Entry of bacteria into blood

- Obstruction causes stasis, impairs washout and raises pelvic pressure. Bacteria and endotoxin then enter the blood by **pyelovenous and pyelolymphatic backflow**.
- High irrigation pressure at endoscopy does the same, and so does fragmenting infected stones, which release endotoxin.
- An obstructed infected kidney is an **abscess with a collecting system**. Antibiotics barely reach it, and bladder urine may be sterile.

2. Host response

```
Endotoxin (LPS, lipopolysaccharide) / Gram-positive wall products
  → TLR4 (Toll-like receptor 4) on macrophages → NF-κB (nuclear factor κB)
  → cytokines TNF-α (tumour necrosis factor), interleukins IL-1, IL-6, IL-8
  → complement + coagulation activation + endothelial injury
```

- **Vasodilatation:** inducible nitric oxide causes low systemic vascular resistance (warm shock).
- **Capillary leak:** glycocalyx damage leads to hypovolaemia and tissue oedema.
- **Coagulopathy:** tissue factor release causes microthrombi, platelet consumption and DIC (disseminated intravascular coagulation).
- **Myocardial depression** and **mitochondrial dysfunction:** cells cannot use oxygen, so lactate rises.

- **Immunoparalysis:** a later anti-inflammatory phase predisposes to secondary infection.

3. Organ dysfunction: AKI (acute kidney injury), ARDS (acute respiratory distress syndrome), encephalopathy, liver dysfunction, DIC, multi-organ failure.

Clinical features

- Fever or hypothermia, rigors, flank pain, tachycardia, tachypnoea, confusion, oliguria, hypotension. Often starts within hours of an endourological procedure.

Investigations

- **Lactate**, blood cultures ×2, urine culture (and pelvic urine at drainage), blood count, creatinine, bilirubin, platelets, coagulation, arterial gas, glucose. Procalcitonin helps decide when to stop antibiotics.
- **Imaging:** bedside ultrasound (hydronephrosis, debris); **CT (computed tomography)** for stone, gas, abscess.

Management

A. Resuscitation and antibiotics (Surviving Sepsis Campaign 2021)

```
Lactate + cultures → IV (intravenous) antibiotics within 1 hour
→ 30 mL/kg crystalloid if hypotensive or lactate ≥4 mmol/L
→ noradrenaline for MAP ≥65; add vasopressin
→ refractory shock: hydrocortisone 200 mg/day
→ ICU (intensive care unit); re-measure lactate
```

- **Empirical antibiotics** by local resistance: third-generation cephalosporin or piperacillin–tazobactam ± aminoglycoside; carbapenem if ESBL risk. Adjust for renal function.

B. Source control: the key urological step

- **Drain the obstructed infected kidney urgently:** retrograde **double-J stent** or **percutaneous nephrostomy (PCN)**. Randomised trials show equal effect on sepsis.
  - Stent: suits coagulopathy; avoid high-pressure contrast injection.
  - PCN: better for large or impacted stones, pyonephrosis; allows pelvic urine culture.
- **No definitive stone surgery during sepsis.** Treat the stone after the infection clears.
- Other sources: drain renal or perinephric abscess; emphysematous pyelonephritis (drainage, nephrectomy if diffuse gas); prostatic abscess drainage; change or remove catheter; debride Fournier gangrene.

C. Supportive care

- Oxygen and ventilation, glucose control, transfusion below haemoglobin 7 g/dL, thromboprophylaxis, stress-ulcer prophylaxis, renal replacement if needed.

D. De-escalation

- Narrow antibiotics by culture at 48–72 h. Usual course 7–10 days; longer with an undrained focus. Step down to oral therapy when stable.

Special situations

- **Emphysematous pyelonephritis** (diabetics): fluids, glucose control, antibiotics, drain; nephrectomy for diffuse gas without fluid.
- **Pregnancy:** stent or nephrostomy; avoid fluoroquinolones.
- **Fungal urosepsis:** remove catheter, antifungal, drain fungal balls.
- **Prognosis:** urosepsis does better than sepsis from other sites if the source is drained early; delay in drainage is the main avoidable cause of death.

Prevention

- Sterile urine before elective endourology; antibiotic prophylaxis; low irrigation pressure and access sheath; short operating time with infected stones; stage the procedure if pus is found.

Diagram to draw: obstructed kidney leading to backflow, cytokine cascade and shock.

> **Key points to remember:**
>
> - Sepsis = SOFA ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2.
> - Obstruction plus infection is the emergency; bladder urine may be sterile.
> - Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline.
> - Stent and nephrostomy equally effective; delay definitive stone surgery.

Q3a. Complications of intravenous contrast media.

Type: SAQ (10 marks)

Introduction

- Agents are tri-iodinated benzene rings, filtered by the glomerulus. Modern practice uses **non-ionic low-osmolar** (iohexol, about 600–850 mOsm/kg) or **iso-osmolar** (iodixanol, 290) agents. High-osmolar ionic agents caused far more reactions.

Complications

Type	Features
Immediate hypersensitivity (within 1 hour)	Urticaria, angio-oedema, bronchospasm, hypotension, anaphylaxis. Mostly non-IgE (immunoglobulin E) mast-cell release. Severe: 0.22% ionic vs 0.04% non-ionic
Non-immediate (hours to days)	Maculopapular rash; T-cell mediated
Physiological	Warmth, metallic taste, nausea, vasovagal reaction, arrhythmia, seizures
CA-AKI (contrast-associated acute kidney injury)	Creatinine rise ≥0.3 mg/dL in 48 h or ≥1.5 × baseline. Afferent vasoconstriction causes medullary hypoxia, plus direct tubular toxicity. Real risk mainly at eGFR (estimated glomerular filtration rate) <30
Metformin	Lactic acidosis only if AKI develops
Thyroid	Hyperthyroidism in Graves' or nodular goitre; blocks radio-iodine for weeks
Extravasation	Usually minor; rarely compartment syndrome

- **Risk factors:** previous reaction (strongest), asthma, atopy, cardiac disease; for CA-AKI: CKD (chronic kidney disease), diabetes, volume, intra-arterial route. **Shellfish allergy is not a risk.**

Prevention

- Use the lowest dose of low- or iso-osmolar agent; consider ultrasound, MRI (magnetic resonance imaging) or non-contrast CT (computed tomography).
- **Premedication** (prior reaction): prednisone 50 mg at 13, 7 and 1 h, or methylprednisolone 32 mg at 12 and 2 h, plus diphenhydramine 50 mg 1 h before. Breakthrough reactions still occur.
- **CA-AKI:** IV (intravenous) isotonic saline for eGFR <30. **PRESERVE (2018):** bicarbonate and NAC (N-acetylcysteine) no benefit.
- **Metformin:** withhold 48 h only if eGFR <30 or AKI.

Treatment of reactions

```
Stop injection → airway, oxygen, help
Urticaria → observe ± antihistamine
Bronchospasm → salbutamol inhaler
Anaphylaxis → IM (intramuscular) adrenaline 0.3–0.5 mg (1 mg/mL), repeat 5–15 min; IV fluids
Vasovagal → legs up, fluids, atropine
```

Diagram to draw: monomer versus dimer on an osmolality scale.

Key points to remember:

- Non-ionic agents: severe reactions 0.04% vs 0.22% ionic.
- Anaphylaxis: IM adrenaline 0.3–0.5 mg.
- CA-AKI: saline if eGFR <30; bicarbonate and NAC useless.

Q3b. Flexible Ureterorenoscopy

Type: SAQ (10 marks)

Introduction

- Flexible ureterorenoscopy (fURS) or RIRS (retrograde intrarenal surgery) inspects and treats the whole upper tract retrogradely. Marshall reported it in 1964; active deflection and a working channel made it practical.

Instrument

Feature	Details
Tip	4.5–7.5 Fr, tapered (digital tips up to about 8.5 Fr)
Working channel	3.6 Fr (laser fibre, basket)
Deflection	≥270° both ways; lower pole needs about 140° (up to 175°)
Types	**Fibreoptic** (smallest, image degrades); **digital** (chip on tip, better image, not more durable); **single-use** (LithoVue 2016, 7.7 Fr; cheaper below about 99 cases a year)

- Repair is needed about every 15 cases.

Indications

- **Stones:** renal <2 cm; lower pole <1 cm (or >1 cm as an alternative to PCNL, percutaneous nephrolithotomy); patients who cannot stop anticoagulants; failed SWL (shock wave lithotripsy); obesity.
- **Others:** diagnosis and ablation of low-grade UTUC (upper tract urothelial carcinoma); filling defects, positive cytology; endopyelotomy, short strictures; calyceal diverticulum; essential haematuria; foreign bodies.

Technique

1. Sterile urine; single-dose antibiotic.
2. Cystoscopy; **safety guidewire**; retrograde pyelogram.
3. **Access sheath** (usually 12/14 Fr) over a stiff wire under fluoroscopy: lowers intrarenal pressure, allows repeated passes; severe ureteric injury in about 1 in 8.
4. Low-pressure irrigation; **Ho:YAG (holmium:yttrium-aluminium-garnet) or thulium fibre laser**: dusting or fragment-and-basket. Keep power ≤10 W in ureter, ≤25 W in kidney (tissue damage above 43 °C).
5. Inspect ureter on withdrawal; stent unless uncomplicated.

- **Difficult lower pole:** steep infundibulopelvic angle, long (>10 mm) or narrow (<5 mm) infundibulum (EAU (European Association of Urology) factors against SWL; they also hinder fURS). Relocate the stone to an upper calyx with a basket.

Complications

- Sepsis about 5%; stricture about 3%; bleeding or subcapsular haematoma <1%; perforation; avulsion <1%; stent symptoms; scope damage.

Recent advances

- Suction access sheaths, intrarenal pressure sensors, thulium laser, robotic fURS.

Diagram to draw: flexible scope deflected into the lower pole calyx through an access sheath.

> **Key points to remember:**
> - 3.6 Fr channel; ≥270° deflection.
> - Access sheath lowers pressure but injures ureter in about 1 in 8.
> - Laser power ≤10 W ureter, ≤25 W kidney.

Q3c. Perinephric abscess

Type: SAQ (10 marks)

Definition

- Pus **outside the renal capsule but within Gerota fascia**. Rupture through Gerota fascia gives a **paranephric** abscess.

Aetiology

- Rupture of a renal cortical abscess or pyonephrosis (commonest); extravasation of infected urine from obstruction (stone).
- **Haematogenous** spread in about a third (skin source, *S. aureus*).
- Infected perirenal haematoma or urinoma (trauma, surgery).
- Spread from bowel (Crohn disease, perforation), pancreas, spine.
- Risk factors: **diabetes (about a third)**, stones, obstruction, polycystic kidneys on dialysis.
- Organisms: *E. coli*, *Proteus*, *S. aureus*; anaerobes with bowel sources.

Clinical features

- **Insidious:** symptoms >5 days in most; more than a third afebrile.
- Flank pain, tenderness, flank mass (about half); psoas sign (flexed, externally rotated hip).
- **Suspect it when "pyelonephritis" persists:** pyelonephritis defervesces within 4–5 days; perinephric abscess does not.

	Pyelonephritis	Perinephric abscess
Symptoms before admission	≤5 days	>5 days
Fever on antibiotics	Never >4 days	≥5 days (median 7)

Investigations

- Leucocytosis, raised creatinine, pyuria; urine culture identifies the organism in only about a third; blood cultures.
- **CT (computed tomography): investigation of choice.** It shows the collection, its spread (psoas, flank), the cause (stone, obstruction) and plans drainage.
- Ultrasound: quick; guides drainage. Plain X-ray: loss of psoas shadow, scoliosis.

Management

- **Antibiotics immediately:** aminoglycoside plus antistaphylococcal penicillin; cephalosporin or vancomycin if allergic. Adjust to culture.
- **Drainage:** small (<3 cm) in an immunocompetent patient may resolve with antibiotics. Larger or non-responding: **CT- or ultrasound-guided percutaneous drainage** (first line); open drainage for loculated or failed cases.
- **Treat the cause:** relieve obstruction (stent or nephrostomy); **nephrectomy** for a non-functioning, pyonephrotic kidney, at the same time or staged.
- Mortality fell from 40–50% to about 12% with CT; delay in diagnosis causes most deaths.

Diagram to draw: coronal kidney showing renal, perinephric and paranephric spaces.

> **Key points to remember:**
> - Within Gerota fascia; diabetes in a third.
> - Fever beyond 4–5 days of antibiotics: get a CT.
> - Percutaneous drainage first line; nephrectomy if non-functioning.

Q3d. Ciprofloxacin

Type: SAQ (10 marks)

Class and mechanism

- Second-generation **fluoroquinolone**, derived from nalidixic acid.
- Inhibits **DNA gyrase** (topoisomerase II, main target in Gram-negatives) and **topoisomerase IV** (Gram-positives). This blocks DNA supercoiling and replication: **bactericidal**.

Spectrum

- Enterobacteriaceae, ***Pseudomonas aeruginosa***, *Haemophilus*, *S. saprophyticus*, *S. aureus* (methicillin-sensitive); *Legionella*, some mycobacteria.
- Weak against streptococci and enterococci; **no anaerobic cover**, so bowel and vaginal flora are partly spared.

Pharmacokinetics

- Oral bioavailability about 70%; IV (intravenous) and oral forms; half-life about 4 h.
- **Concentration-dependent killing:** AUC/MIC (area under curve to minimum inhibitory concentration) predicts effect; post-antibiotic effect.
- High urine and tissue levels; **good prostatic penetration**.
- Renal excretion: reduce dose if creatinine clearance <30 mL/min.

Doses

- Cystitis 250 mg twice daily × 3 days; pyelonephritis 500 mg twice daily × 7 days (or 1 g extended-release daily); IV 400 mg 12-hourly.
- Chronic bacterial prostatitis: 4–6 weeks.

Indications and restrictions

- Pyelonephritis and complicated UTI (urinary tract infection) **only where local resistance ≤10%**; prostatitis; prophylaxis for transrectal biopsy (falling, because of resistance).
- **Not first line for uncomplicated cystitis** (FDA (US Food and Drug Administration) 2016; European restrictions 2019): use only when no alternative.

Resistance

- *gyrA*/*parC* mutations, efflux pumps, reduced porins, plasmid *qnr* genes. Linked to ESBL (extended-spectrum β-lactamase) *E. coli* (ST131). Main cause of post-biopsy sepsis.

Adverse effects

- GI (gastrointestinal) upset (commonest), photosensitivity, *C. difficile*.
- **FDA boxed warnings:** tendinopathy and **Achilles rupture** (2008; risk higher with age, steroids, renal failure); peripheral neuropathy (2013); disabling effects (2016); psychiatric effects and dysglycaemia (2018); aortic aneurysm and dissection (2018).
- CNS (central nervous system): seizures, confusion; QT prolongation.

Interactions

- Antacids, iron, zinc, calcium block absorption (separate by ≥2 h). Raises theophylline and tizanidine levels; potentiates warfarin.

Pregnancy and children

- Avoid: **cartilage damage** in immature animals. Also avoid in breastfeeding.

Key points to remember:

- DNA gyrase and topoisomerase IV; concentration-dependent.
- Use for pyelonephritis only if local resistance ≤10%.
- Tendon rupture, neuropathy, aortic aneurysm; avoid in pregnancy and children.

Q3e. Hassan's canula.

Type: SAQ (10 marks)

Assumed to mean the Hasson cannula, used for open laparoscopic access.

Introduction

- Described by **Harrith Hasson (1971)** for open ("direct vision") laparoscopy. The first port is placed under vision instead of a blind Veress needle and trocar.
- **Parts:** outer sleeve with a cone-shaped (olive) adapter, **blunt obturator**, struts to anchor fascial stay sutures, and a gas inlet with tap.

Indications

- **Previous abdominal surgery or expected adhesions** (main indication).
- Organomegaly, pregnancy, aortic aneurysm, children, thin patients.
- Retroperitoneoscopy and extraperitoneal access (open start, then balloon dilatation).

Technique (transperitoneal)

1. 2-cm semicircular infra-umbilical incision.
2. Open fascia and then peritoneum separately, under vision, finger-sized.
3. Confirm entry by sight and finger sweep (free adhesions).
4. Place two 0 silk fascial stay sutures.
5. Insert the cannula with its **blunt obturator**.
6. Seat the cone in the defect; wrap stay sutures round the struts to seal.
7. Remove obturator; insufflate at high flow (to 12–15 mmHg).
8. At closure, tie the stay sutures to close the fascia.

- **Retroperitoneal:** 2–2.5 cm incision below the 12th rib tip, finger dissection onto psoas, balloon dilatation, then port.

Advantages over Veress

- No blind puncture: fewer major vessel and bowel injuries; no preperitoneal insufflation; no gas embolism from intravascular insufflation.
- Fast insufflation; easy fascial closure (less port-site hernia).

Disadvantages

- Larger incision; **gas leak** around the port; slightly longer in obese patients.
- Early retroperitoneal use caused subcutaneous emphysema and hypercarbia, now solved by balloon-cuffed blunt ports.

Complications

- Bowel injury if adherent under the wound (repair through the same incision); wound infection; port-site hernia; vascular injury is still possible.

Evidence

- Reviews show no single entry method is clearly safest for major injury; open entry has fewer failed entries.

Diagram to draw: Hasson cannula with cone, stay sutures and blunt obturator in the abdominal wall.

Key points to remember:

- Hasson 1971: open entry under vision.
- Main indication: previous surgery or adhesions.
- Avoids blind Veress injuries; drawback is gas leak.

Q3f. Proteinuria

Type: SAQ (10 marks)

Definition

- Normal protein excretion is **<150 mg/day** (albumin <30 mg/day). The filter blocks albumin and the proximal tubule reabsorbs filtered small proteins.
- Normal mix: about 30% albumin, 30% globulins, 40% tissue proteins (Tamm–Horsfall).
- **Nephrotic range:** >3.5 g/day, with hypoalbuminaemia, oedema and hyperlipidaemia.

Quantification (KDIGO, Kidney Disease: Improving Global Outcomes)

Category	ACR (albumin-to-creatinine ratio)	PCR (protein-to-creatinine ratio)	24-h protein
A1 normal–mild	<30 mg/g (<3 mg/mmol)	<150 mg/g (<15 mg/mmol)	<150 mg
A2 moderate	30–300 mg/g	150–500 mg/g	150–500 mg
A3 severe	>300 mg/g (>30 mg/mmol)	>500 mg/g	>500 mg

- **Spot early-morning ACR is first line**; PCR if non-albumin protein suspected; 24-h urine with creatinine check.
- **Dipstick** (tetrabromophenol blue) detects albumin only, from 20–30 mg/dL; misses light chains. **Sulfosalicylic acid** detects all proteins.

Types

Type	Mechanism	Amount	Example
Glomerular (commonest)	Leaky filter	Often >2 g; albumin ≈70%	Diabetes, glomerulonephritis, FSGS (focal segmental glomerulosclerosis)
Tubular	Failed reabsorption of small proteins	<2 g; mainly globulins	Fanconi syndrome, heavy metals, drugs
Overflow	Excess plasma protein	Variable	Myeloma (Bence Jones), myoglobin

- **Transient:** fever, exercise, heart failure. **Orthostatic:** young men, upright only, <1 g; benign. Test with a split (supine overnight vs upright day) collection.

Evaluation

```
Positive dipstick → repeat; exclude UTI (urinary tract infection), fever, exercise
→ persistent → ACR/PCR or 24-h protein
→ creatinine/eGFR (estimated glomerular filtration rate), sediment (dysmorphic red cells,
casts), ultrasound
→ glomerular pattern → nephrology ± renal biopsy
→ globulins → immunofixation (myeloma)
```

Urological relevance

- **Haematuria work-up:** proteinuria, casts or dysmorphic cells suggest medical renal disease; urological bleeding rarely raises protein much.
- **Reflux nephropathy:** proteinuria marks progression to renal failure.
- **After nephrectomy or partial nephrectomy:** >50% loss of mass causes hyperfiltration, proteinuria and FSGS. AUA (American Urological Association) 2021 counts proteinuria as a reason to prefer nephron sparing and refer to nephrology.
- **Living donors:** significant proteinuria excludes donation.

Management

- Treat the cause; blood pressure control; **ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)**; SGLT2 (sodium–glucose cotransporter-2) inhibitors.

Key points to remember:

- Normal <150 mg/day; nephrotic >3.5 g/day.
- ACR A2 30–300, A3 >300 mg/g.
- Dipstick misses light chains; orthostatic proteinuria is benign.

Rapid Revision Sheet – 2017-Summer Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss anatomy of kidney. Describe role of nephron sparing surgery in small renal mass. (LAQ)

- Five segmental end arteries; Brödel line avascular; puncture at the fornix.
- SRM ≤4 cm: 20–30% benign; PN is standard, ablation <3 cm, surveillance <2 cm.
- R.E.N.A.L. 4–6 / 7–9 / 10–12; trifecta includes warm ischaemia ≤25 min.
- EORTC 30904: PN better function, similar cancer control.

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- Sepsis = SOFA ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2.
- Obstruction plus infection is the emergency; bladder urine may be sterile.
- Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline.
- Stent and nephrostomy equally effective; delay definitive stone surgery.

Q3a. Complications of intravenous contrast media. (SAQ)

- Non-ionic agents: severe reactions 0.04% vs 0.22% ionic.
- Anaphylaxis: IM adrenaline 0.3–0.5 mg.
- CA-AKI: saline if eGFR <30; bicarbonate and NAC useless.

Q3b. Flexible Ureterorenoscopy (SAQ)

- 3.6 Fr channel; ≥270° deflection.
- Access sheath lowers pressure but injures ureter in about 1 in 8.
- Laser power ≤10 W ureter, ≤25 W kidney.

Q3c. Perinephric abscess (SAQ)

- Within Gerota fascia; diabetes in a third.
- Fever beyond 4–5 days of antibiotics: get a CT.
- Percutaneous drainage first line; nephrectomy if non-functioning.

Q3d. Ciprofloxacin (SAQ)

- DNA gyrase and topoisomerase IV; concentration-dependent.
- Use for pyelonephritis only if local resistance ≤10%.
- Tendon rupture, neuropathy, aortic aneurysm; avoid in pregnancy and children.

Q3e. Hassan's canula. (SAQ)

- Hasson 1971: open entry under vision.
- Main indication: previous surgery or adhesions.
- Avoids blind Veress injuries; drawback is gas leak.

Q3f. Proteinuria (SAQ)

- Normal <150 mg/day; nephrotic >3.5 g/day.
- ACR A2 30–300, A3 >300 mg/g.
- Dipstick misses light chains; orthostatic proteinuria is benign.